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Using microsprinkler irrigation to reduce leaching in a shrink/swell clay soil

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Abstract

Preferential flow allows agricultural chemicals to rapidly move past crop root zone to subsoil, potentially contaminating groundwater. Deep shrinkage cracks in a shrink/swell soil may serve as preferential flow channels. Design and operation of an irrigation system is critical in shrink/swell soils in order to reduce or eliminate preferential flow while providing sufficient water for plant growth. The objective of this study was to compare microsprinkler irrigation (MI) and surface flood irrigation (FI) systems for their ability to reduce preferential flow and chemical leaching through a shrink/swell soil in a pear orchard. The MI system applied water at a rate of 2.8 mm h^{-1}, and the FI followed the traditional orchard irrigation method. Bromide tracer was applied to the pear orchard. Soil cores and percolate samples from passive capillary samplers (PCAPS) installed at 1.2 m depth were collected to test for the presence of Br tracer in the soil and percolate. The MI system greatly decreased water and Br leaching from soil to the PCAPS in comparison to FI. The percolate in MI was nearly zero. Most of the loss of Br occurred in the first one or two FI events. MI systems, such as that used in this study should be considered as an alternative to FI in such shrink/swell soils to reduce macropore flow.

Keywords: Shrink/swell soil; Microsprinkler; Preferential flow

1. Introduction

In structured soils, preferential flow occurs through soil voids, such as worm holes, root channels, and shrinkage cracks (Bouma and Dekker, 1978; Beven and Germann,

Abbreviations: FI, flood irrigation; MI, microsprinkler irrigation

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1982; Kneale, 1986; Edwards et al., 1989). Water and solutes are transported through macropores much more rapidly than through the soil matrix. Preferential flow through macropores results in rapid transport of agrichemicals through the biologically active root zone. This reduces the opportunity for degradation of chemicals within the soil, and increases the risk of groundwater contamination by agrichemicals.

Rainfall intensity and irrigation method affect preferential flow through soil. Bouma and Dekker (1978) demonstrated that application rate and total quantity of applied solution containing a dye tracer affected the number and depth of stained bands in a dry clay soil. Preferential flow through macropores begins when rainfall or irrigation rates exceed the soil infiltration rate, or when saturated flow occurs. Edwards et al. (1992) found in a no-till silt loam that a high rainfall rate resulted in greater movement of atrazine through macropores than a low rainfall rate of equal volume.

In shrink/swell soils, deep cracks form in the topsoil as the soil dries and shrinks. If the water application rate exceeds the soil matrix intake rate, as with surface flood irrigation (FI) or high rate sprinkler irrigation, the excess water will flow into the cracks and contribute to macropore flow (Topp and Davis, 1981; Kosmas et al., 1991; Mitchell and van Genuchten, 1993). Poor irrigation management, inadequate scheduling, and poor soil management can combine to result in significant leaching and potential degradation of groundwater quality, regardless of irrigation method. Thus, a specific strategy is needed for selecting and managing an irrigation system in such shrink/swell soils to reduce macropore flow while providing sufficient water for plant growth. Such a strategy has not been developed yet. Some researchers (e.g. Mitchell and van Genuchten, 1993) suggested that the strategy of irrigation in shrink/swell soils should be to take advantage of the rapid water intake rate of dry cracked soil using FI, because sprinkler irrigation may result in the closure of cracks near soil surface which would limit water intake. However, in areas with shallow groundwater table, such FI may cause groundwater contamination by agrichemicals. Selker et al. (1995) reported that macropore flow was eliminated by reducing irrigation rate to 3 mm h^{-1}, using an ultralow rate irrigation device.

The objective of this study was to compare the movement of water and chemicals under MI and surface irrigation. Irrigation method effects on soil available water and pear fruit growth and post-harvest quality have been published in a companion paper (Chen et al., 2001). These studies were part of a larger project investigating the behavior of water and chemical movement through shrink/swell soils used for intensive tree fruit agriculture. Additional results and details of the larger project can also be found in Chen (1998).

2. Materials and methods

The study was conducted in 1995–1996, at the Southern Oregon Research and Extension Center (SOREC) near Medford, Oregon. The soil is classified as Carney Clay (fine, montmorillonitic, Mesic Typic Chromoxererts). This soil is moderately well drained, has very low matrix permeability, and is underlain by a shallow groundwater table that fluctuates seasonally between 0.8 and 4 m below the surface (Fig. 1). The soil has high shrink/swell characteristics (Chen, 1998). The soil is underlain by weathered

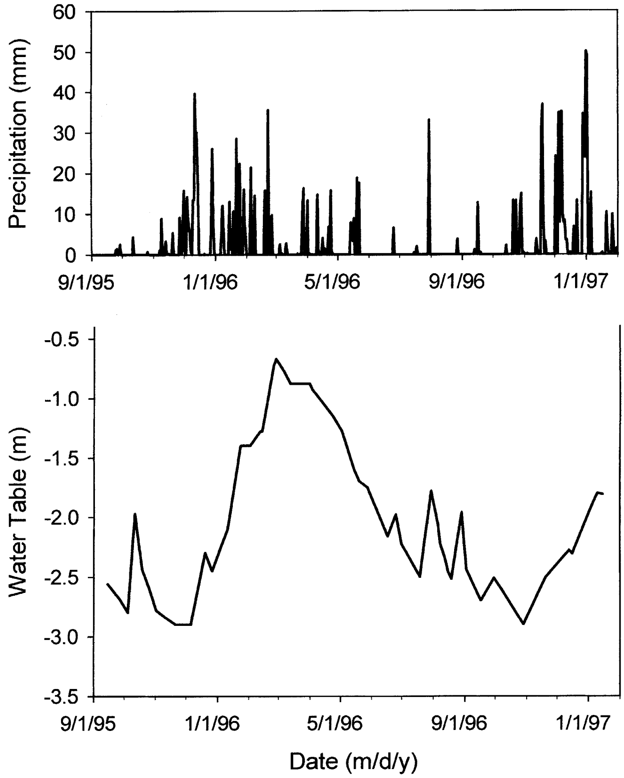

Fig. 1. Precipitation distribution and depth of the groundwater table at experiment site from September 1995 to January 1997.

bedrock at about 1.5 m depth, which contains extensive, permanent cracks and channels. When the soil is dry, soil shrinkage cracks and then channels in the bedrock added together extend to at least 2.5 m depth.

Mean annual precipitation is 479 mm, with approximately 75% occurring between October and March. During the 1995–1996 study period, the precipitation pattern was fairly typical, and included one large thunderstorm on 30 July 1996, that delivered 33 mm water in about 1 h (Fig. 1). The pear trees (*Pyrus communis*, cv. Bosc) in the study orchard were planted in 1935 on a 7.3 m × 7.3 m spacing. Grass strips are maintained between tree rows, but the soil between trees within a row was kept relatively weed-free with herbicides following standard orchard management practices.

The general approach to this study included: (1) establishment of two irrigation systems in the pear orchard; (2) application of Br as a tracer to track water and solute movement in the soil; (3) installation of passive capillary samplers (PCAPS) samplers to allow collection and measurement of water and solutes that had moved beyond the root zone to the PCAPS 1.2 m depth; (4) periodically sampling of soil cores to measure the movement of the remaining Br within the soil column between the soil surface and the PCAPS.

2.1. Irrigation systems

Two irrigation systems, MI and FI, were tested side by side in the pear orchard that previously was surface flood irrigated. For the MI system, two emitters were installed within the tree row between each pair of trees. Emitters were installed at 3.7 m intervals and at a height of 0.25 m. This gave zero overlap of irrigation water. The MI system used Dan 8000 series emitters with 1.4 mm nozzle and 207 kPa pressure (Netafilm, New York), which applied water at a rate of 2.8 mm h^{-1} and wetted an area 3 m in diameter. Uniformity of the microsprinkler was tested and assured by measuring irrigation amounts collected at different locations along the diameter of wetting circle (data not shown). Digital irrigation timers were used to control the irrigation system. During the 1995 irrigation season, the MI system ran 12 h weekly in a single event, and operated 6 h twice per week in 1996. Irrigation treatments began in late July in 1995 following PCAPS installation (see Section 2.2) and irrigation system repair. Irrigation treatments began in late June 1996. Irrigations were scheduled based on previous studies of pear trees done in this orchard, where evapotranspiration was 5 mm per day in July and August (Buchheim and Roseberg, 1994).

Gated pipe was used to deliver water to the surface FI treatments. Discharge into each tree row was 99 l min^{-1}. Assuming completely uniform application, this gave a net application of about 100 mm per day. Irrigation was carried out once every 2–3 weeks. The cumulative irrigation amount applied by each system is shown in Fig. 2, and was essentially equal for both treatments.

2.2. Passive capillary samplers and soil percolate

Three PCAPS were installed at a depth of 1.2 m under each irrigation system to collect percolate. Although pear roots can penetrate beyond 1.2 m depth, previous studies in this orchard showed that pear trees absorb water and nutrients almost exclusively from the upper 1.2 m soil profile, except under extreme drought conditions (Buchheim and Roseberg, 1994). Also, installation at 1.2 m placed the PCAPS water collection surface just above the interface between subsoil and weathered bedrock. Thus, the PCAPS measured water passing through soil only. Two PCAPS were installed in the area between two trees at a distance of 3.7 m from each other. The third PCAPS was installed in the next area between two trees, at 1.8 m from the first tree (Fig. 3). Each PCAPS was constructed of 0.5 cm thick fiberglass, with dimensions of $0.35\,m \times 0.85\,m \times 0.67\,m$. The boxes were divided into three chambers of equal volume, (66 l each). The top of the PCAPS was designed as an intercepting pan with a 1.5 cm tall edge around the perimeter. One hole was drilled over each chamber, and one wick was installed through each hole. The upper unbraided end of the wick was glued to the intercepting pan and the lower braided portion of the wick was allowed to hang down through the drilled hole and into the bottom of the chamber to provide 0–6 kPa tension, which would allow the PCAPS to collect both saturated and unsaturated flows from the soil (Boll et al., 1992; Knutson and Selker, 1994). A withdrawal tube extending to the soil surface was inserted into each chamber to allow withdrawal of water samples from the chambers. More detailed information about PCAPS design, construction, and field application has been published

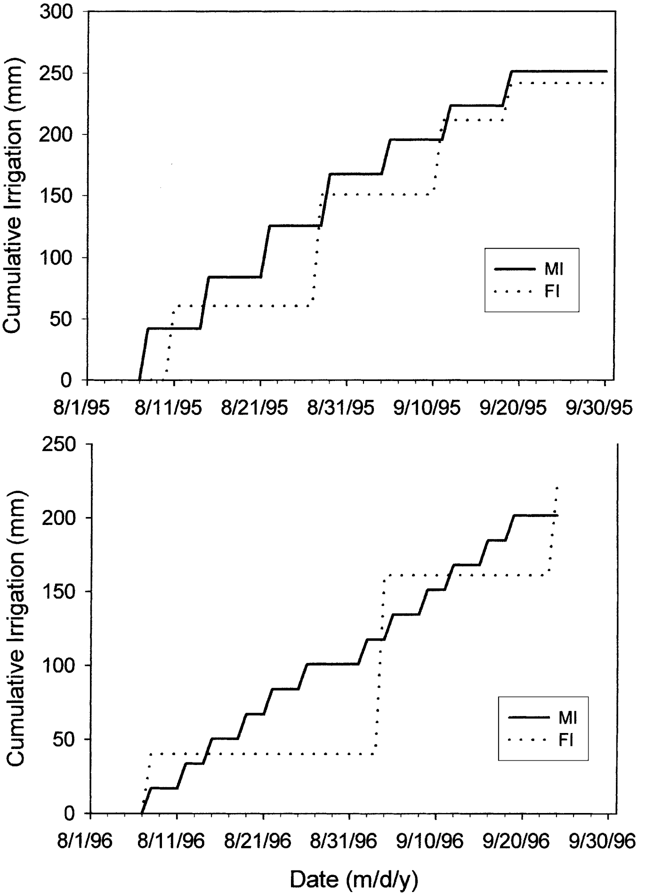

Fig. 2. Cumulative irrigation amount for MI and FI during the growing season of 1995 and 1996.

previously (Boll et al., 1992; Knutson et al., 1993; Knutson and Selker, 1994; Knutson and Selker, 1996; Brandi-Dohrn et al., 1996a,b).

A vacuum pump, which applied 60–90 kPa suction, was used to withdraw water samples from each collection chamber through the access tubes. Samples were taken once every 2 weeks during the summer season.

2.3. Br tracer and soil samples

On 12 July 1996, calcium bromide was dissolved in water and sprayed uniformly in four passes over the surface of a 22 m $\times$ 22 m test area for each irrigation system using a bicycle frame herbicide sprayer. The sprayed area was centered on the PCAPS sites. The tracer was applied at a rate of 18 g m^{-2} as Br, and the application rate of the spray solution was 128 ml m^{-2}.

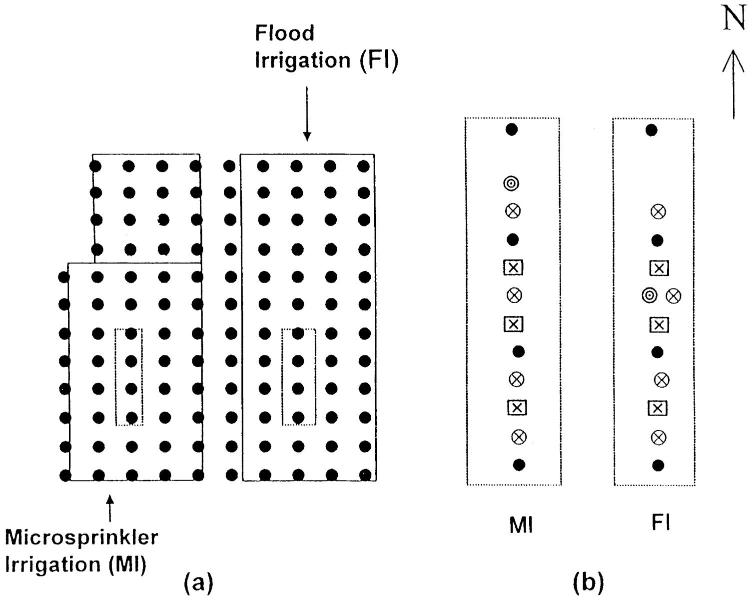

Fig. 3. Schematic of field experiment: (a) layout of irrigation treatments; (b) equipment installation in the test areas. The meaning of the symbols are: (●) pear tree; (⊗) neutron probe access tube; (⊠) PCAPS; (◎) groundwater observation well; (▭) irrigation area; (▭) tracer application and sampling area.

To investigate the movement and distribution of water and solute in the soil profile, soil core samples were taken on 21 September 1995 (background level before the tracer application), on 5 August 1996 (after the first thunderstorm following the tracer application), and on 8 October 1996 (after the pear harvest). Four soil cores were taken from each test area under each irrigation system. Core samples were taken at 2 m distance from one another along the tree rows using a Lord Soil Sampler #225 (Soil Moisture Equipment Corp., Santa Barbara, CA) at 15 cm intervals down to a depth of 120 cm. Samples were air dried and ground to pass a 2 mm sieve. Bromide was extracted from soil samples with water, using a soil to water ratio of 1:2 by mass. Samples were shaken for 30 min, then centrifuged.

Bromide in soil extractions and PCAPS water samples was measured using a Model 94-35 bromide electrode (Orion Research Inc., Boston, MA). The sensitivity of the electrode and the recovery rate of Br in soil were tested using a series of spiked soil samples in the lab. Background Br in the experimental soil was less than 2 mg Br kg^{-1}Soil and the recovery in spiked samples was over 94%.

3. Results and discussion

PCAPS samplers were used to measure water and solutes movement beyond the root zone, 1.2 m depth, in a way that was more quantitative than other sampling devices, such as suction cup samplers (Brandi-Dohrn et al., 1996a,b). These measurements provide a strong indication of the existence and timing of preferential flow of water and solutes. Periodic soil core samples were obtained to measure the movement of the remaining Br within the soil column, between the soil surface and the PCAPS. This allowed a periodic snapshot of Br movement within the profile, and would help explain the movement of solutes via non-preferential (piston) flow through the soil matrix.

3.1. Volume of percolate and Br tracer in PCAPS

Cumulative volume of percolate collected in each of the PCAPS under FI was much larger than under MI (Fig. 4). The percolate collected under MI was almost zero. There

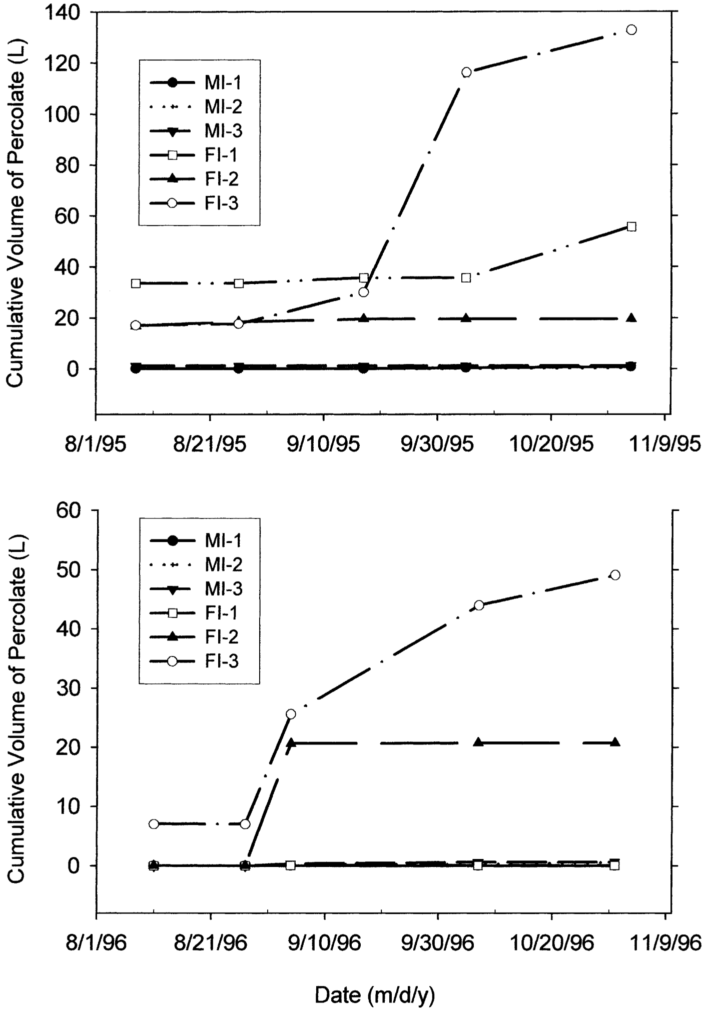

Fig. 4. Cumulative volume of percolate collected in PCAPS under MI and FI during the growing season of 1995 and 1996.

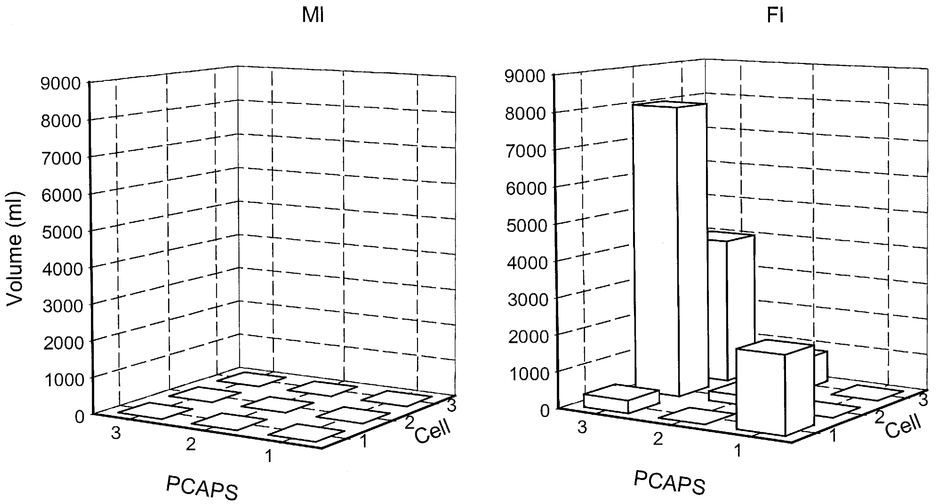

Fig. 5. Volume of percolate collected on 17 September 1995 for each chamber of the three PCAPS under MI and FI.

was great variation among PCAPS in the volume of percolate collected (Fig. 4) and among the chambers within each PCAPS (Fig. 5) under FI. Furthermore, some PCAPS chambers consistently received larger percolate volumes than others (e.g. FI-3). This indicates the existence of ongoing connections of those chambers to preferential flow channels. The amount of Br collected in PCAPS was also much greater under FI than under MI (Table 1). Similarly, during the period between August and October of 1995 and 1996, the mean volumes of percolate collected from three PCAPS were much greater in FI than in MI, although the total amount of irrigation during this period under the two systems was similar (Table 1), assuming uniform application of water in FI.

3.2. Uniformity of flood irrigation

The experimental area was selected, in part, for its fairly uniform slope (about 0.5%) and uniformity of management. For the scale of the experimental orchard, the application

Table 1
Water applied and mean of percolate and tracers collected from three PCAPS

	8/8/1995 to 10/5/1995		8/8/1996 to 10/7/1996	
	MI	FI	MI	FI
PP[a] (mm)	5.6	5.6	18.5	18.5
IR[b] (mm)	223.4	241.9	210.0	221.7
Percolate (mm)	1.5 (±2.0)[c]	190.0 (±172.6)	1.0 (±0.7)	113.9 (±40.2)
Br (g m^{-2})	–	–	0.01 (±0.01)	1.76 (±0.34)

[a] PP: precipitation.
[b] IR: irrigation.
[c] The number in the parenthesis is one standard deviation of volumes from three PCAPS.

of FI water could be non-uniform because of shrinkage cracks and locally uneven orchard ground. Since the clay soil shrank when it dried, forming deep cracks, surface FI water was observed to quickly flow into cracks. This initial, rapid, preferential flow could potentially reach the groundwater, if shrinkage cracks connected to deep preferential channels, such as dead root channels. Due to this flow into cracks, lateral movement of the FI water over the orchard surface was slow. To improve uniformity, the orchard manager sometimes moved the pipe several times during individual irrigation events to decrease the time required for the irrigation water to cover the whole orchard. This was a change from the normal irrigation scheduling, but one which improved uniformity of application. In addition, because of repeat tractor traffic across the orchard, two tire tracks tended to occur through the inter-row grass strip in the orchard. Thus, when surface flood water was applied, the water flowed down the orchard along the tire tracks more rapidly than elsewhere, which initially resulted in non-uniform wetting of the surface soil. Once the orchard was fully flooded, this effect was clearly less prominent than it was when irrigation was begun.

In the measurement area, however, the water application was more uniform than it was for the whole orchard, because there was no tractor track over the measurement area. Furthermore, the travel time for the surface irrigation water passing across the measurement area was relatively short, even though there were shrinkage cracks in the topsoil. Therefore, rather than caused by non-uniformly application of surface irrigation water, the variation of percolate volumes collected in PCAPS was more likely caused by spatial variability of macropores and connection of PCAPS to macropore networks. In all cases, irrigation water covered the FI treatment area by a point in time that was between one-fourth and one-half of each irrigation event's total run time. However, other than visual observation there was no measurement conducted to evaluate uniformity of surface FI.

3.3. Br tracer recovered from soil samples

The mean mass of Br in four soil cores recovered from the 120 cm soil profile was significantly less for FI than MI on 5 August and 8 October 1996 ($P < 0.10$, Table 2). Most of the Br loss from the soil under FI occurred in the first two irrigation events. During the period from the time Br was applied on 12 July 1996 to 5 August 1996, there were two irrigation events and one thunderstorm that delivered 215 mm water in total. The Br lost from the 120 cm soil profile under FI was 43% of that applied. However,

Table 2
Mean bromide tracer recovered in top 120 cm of soil from four soil cores

	8/5/1996		10/8/1996	
	Br (g m^{-2})	Applied Br (%)	Br (g m^{-2})	Applied Br (%)
MI	17.2 a[a]	96	15.4 a	86
FI	10.3 b	57	9.5 b	53

[a] Mean values within a column followed by different letters are significantly different at $P < 0.10$ using Student *t*-test.

during the period from 5 August to 8 October, when there were three irrigation events applying 240 mm of water, the loss of Br from the 120 cm soil was only 4%.

The reason for the great amount of Br initially washing out of the soil under FI probably was because, under FI, water can flow quickly down cracks. In that case, there would be a shorter time for irrigation water to interact with the surface soil matrix, and a larger proportion of water would contribute to macropore flow, taking the soluble Br with it. This would result in a reduced amount of Br moving into the soil matrix. According to Mitchell and van Genuchten (1993), shrinkage cracks took up 63–74% of FI water in a short period of time at the beginning of irrigation, indicating such a scenario was likely for FI in this case. For the low application rate MI irrigation, however, there were longer wetting periods and a larger proportion of water in contact with the soil matrix, as compared to FI irrigation. Therefore, there was a larger proportion of water and Br transported through the soil matrix in a non-preferential (piston-like) flow path for MI.

Irrigation water movement into the soil matrix under MI, could have also helped to prevent rapid leaching of Br by rainfall. Although a thunderstorm on July 30 1996 delivered a significant amount of water (33 mm) in only 1 h (Fig. 1), the Br lost from the soil was minimum (0–4%) with the treatment in MI (Table 2). This suggests that the low intensity irrigation prior to the thunderstorm moved the Br into the soil matrix. Shipitalo et al. (1990) reported that a light rain following chemical application helped to move the chemical into the soil matrix, thereby reducing chemical leaching due to macropore flow induced by subsequent storms. In contrast, the substantial amount of Br lost from FI in the first two irrigation events is consistent with the notion that when FI was used, the Br on the soil surface was more easily mobilized and leached.

Measured field-average Br content profiles may be used as an indicator of average chemical movement in a field scale (Bronswijk et al., 1995). The measured field-average Br content profiles for each irrigation method are shown in Fig. 6. Each data point in

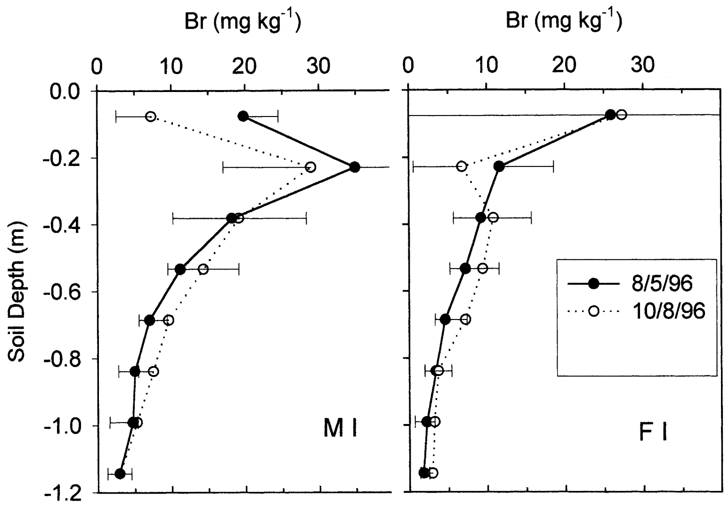

Fig. 6. Distribution of Br tracer in soil profile on 5 August and 8 October 1996 under MI and FI. Each error bar in the graph represents 1 S.E. of four soil samples collected at the same depth.

Fig. 6 represents the average of four soil cores. Comparison of the data collected on the two sampling dates (5 August and 8 October 1996) for the microsprinkler treatment, shows that the maximum concentration peaks for MI had moved downward into the soil profile. In contrast, the downward movement of maximum concentration of Br in the surface FI treatment was not as obvious as in MI treatment. Data of the field-average Br profile for FI showed that there was relatively higher Br concentration in the top 10 cm of the soil profile, but there were also large variations in that layer (Fig. 6). Nevertheless, the front of the Br distribution plume had reached a depth of 80–100 cm of the soil profiles in both irrigation systems. The differences of Br distribution in soil profile between two irrigation treatments are likely due to different flow patterns. As discussed in the previous paragraphs, a larger portion of irrigation water moved through soil matrix in MI treatment than in FL treatment. There were large variations among the soil cores both in MI and FL treatments (Fig. 6).

3.4. Evaluation of PCAPS and soil core Br data

We were not able to close the mass balance for Br between that lost from soil and recovered from PCAPS (Table 1 and 2). During the period between 5 August and 8 October 1996, the mean of Br mass recovered from three PCAPS under FI (1.76 g m^{-2}) was more than the mean mass lost from soil (0.8 g m^{-2}). In contrast, less Br was recovered in PCAPS (0.01 g m^{-2}) than lost from soil (1.8 g m^{-2}) under MI irrigation. This lack of closure could be due to spatial variation of samples and lateral flow. Some lateral flow channels might have been captured by PCAPS and others might have bypassed PCAPS. For example, during the period from 8 August to 7 October 1996, one PCAPS in FI collected 33,900 ml in one chamber, while the other two chambers received only 2000 ml. This indicated that the first chamber was connected to macropore channels and received water more rapidly than the other two.

Under-recovery of Br in PCAPS has also been found by Brandi-Dohrn et al. (1996b) who studied Br movement in a 0.9 ha area of a Willamette silt loam soil using 32 PCAPS over a 2 years period. They found mass recovery of Br within PCAPS ranged from 5.3 to 76.6%, with a mean of 29%. They concluded that the under-sampling of Br was due to the lateral flow bypassing the PCAPS. They suggested that in the 0.9 ha area, the number of PCAPS required to estimate the mean Br concentration with a 30% bound at the 0.05 confidence level is 25, as estimated from the variance between the samplers. Based on this estimation, the number of PCAPS required for the 22 m $\times$ 22 m test area in this study would be 1.4 assuming the same variations between the samplers. However, unlike the soil studied by Brandi-Dohrn et al. (1996b), the soil in this study had shrinkage cracks and was highly structured. There have been no studies to date exploring how many and what design of PCAPS would be appropriate for such highly shrink/swell clay soils, and the Brandi-Dohrn et al. (1996b) study was not complete at the time this experiment was installed. Based on the variations of the volume of percolate collected in the three PCAPS under FI in this study from 8 August to 7 October 1996, twenty-two PCAPS would be required for the 22 m $\times$ 22 m area to estimate mean volume with a 30% bound at the 0.05 confidence level. Unfortunately, installing anywhere near this number of PCAPS in such a clay soil orchard would be very costly to install (approximately US$ 1000 per PCAPS), and destructive to the orchard. Therefore, in hindsight it is clear that use of PCAPS in such cracked-clay pear orchards may need further evaluation.

3.5. Further implications of results

It was seen that surface FI had some advantage for rapid infiltration of water into the root zone through shrinkage cracks, as was also suggested by Mitchell and van Genuchten (1993). In this study, although surface flood and MI methods resulted in similar pear fruit growth, yield, and post-harvest quality (Chen et al., 2001), FI resulted in more water and chemical being leached out of the root zone (1.2 m) with similar amounts of irrigation water applied. Therefore, surface FI exhibited a higher risk of groundwater contamination. This would especially be true in places, such as the pear orchard in this study, where there is a shallow groundwater table and deep continuous preferential channels extending past the rooting depth. Although there would sometimes be a concern of potential soil salinization using MI in arid areas, we did not find salt accumulation in the study orchard near Medford. The reason is that the winter rainfall in the Medford area is enough to annually leach any accumulations of soluble salts out of the root zone. In addition, irrigation water quality is such that salt accumulations are rare and unexpected in the Medford area (Roseberg, 2000; personal communication).

Soil moisture data and solution collected in PCAPS during the winter season of 1995–1996 indicated that the wetting front from rainfall had connected to the groundwater table (Chen, 1998). Although the shallow groundwater table did not affect the sample collection during the summer irrigation season (water table was below the PCAPS), we could not determine the remaining Br movement through the soil matrix to PCAPS because of flood of PCAPS by groundwater during the rainy winter season (water table rose above the PCAPS). Therefore, to prevent groundwater pollution during the rainy season, caution would also need to be taken to minimize the amount of residual nitrogen or soluble pesticide in the orchard after harvest.

4. Conclusions

Given equivalent amounts of applied irrigation, conventional FI resulted in greater loss of Br tracer from the soil profile, and a larger volume of water reaching the PCAPS at 1.2 m depth, than the MI treatment. Low rate microirrigation systems can greatly reduce the amount of water and chemical movement through preferential flow paths. Although FI has advantages when rapid water infiltration into a shrink/swell soil and leaching soluble salts out the root zone are desirable, microsprinkler should be seriously considered as an alternative irrigation system in places where the groundwater is sensitive to contaminant loading and where preferential flow channels extend past the rooting depth. In addition, these systems should allow growers to reduce fertilizer and other chemical applications by retaining a greater fraction of active ingredients within the root zone, in the soil matrix, where they are less likely to be removed by preferential flow.

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